EXPERIMENT - 13

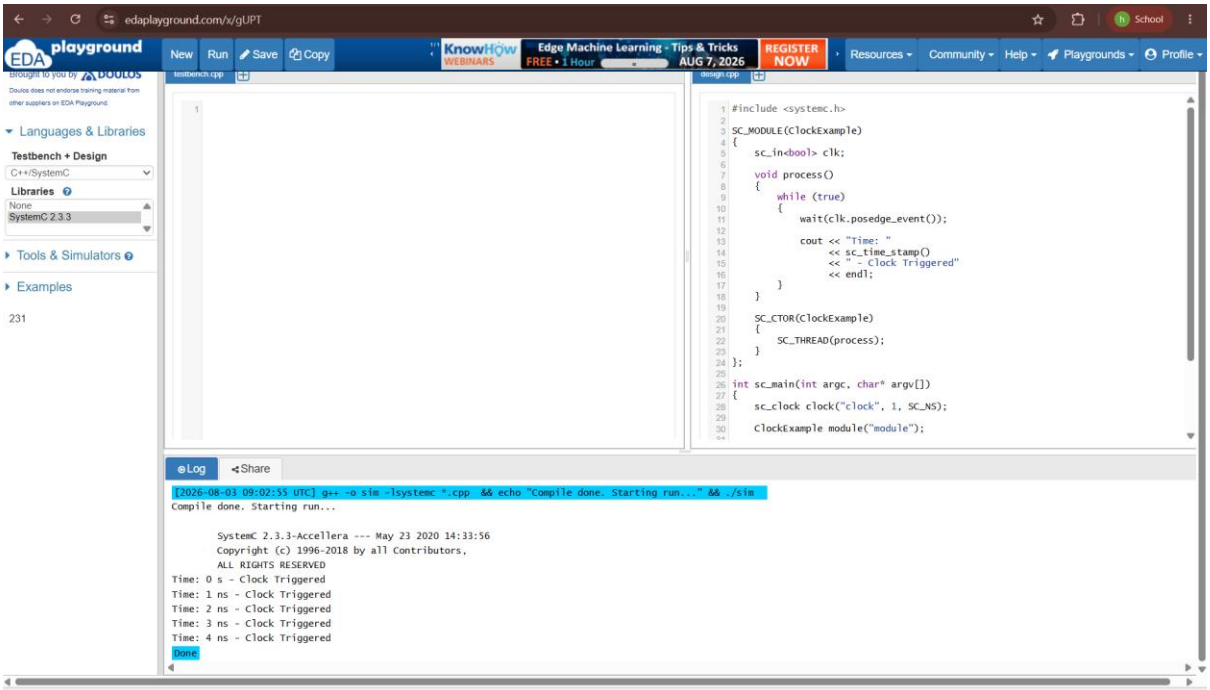
MODELING CLOCK-DRIVEN CONCURRENT PROCESSES USING SC_THREAD AND WAIT() IN SYSTEMC

CODE:

```
#include <systemc.h>
SC_MODULE(ClockExample){
   sc_in<bool> clk;
   void process() {
      while (true){
         wait(clk.posedge_event());
         cout << "Time: "
            << sc_time_stamp()
            << " - Clock Triggered"
            << endl;
      }
   }
   SC_CTOR(ClockExample){
      SC_THREAD(process);
   }
};
int sc_main(int argc, char* argv[]){
   sc_clock clock("clock", 1, SC_NS);
   ClockExample module("module");
   module.clk(clock);
   sc_start(5, SC_NS);
```

```
    return 0;
}
```

SIMULATION:



OUTPUT: